

HARSHVARDHAN GALANDE

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Summary

AI/ML Engineer specializing in LLMs, RAG, and Agentic AI, with hands-on experience building end-to-end AI applications using Python, FastAPI, Qdrant, embeddings, and LLMs. Strong foundation in Machine Learning, NLP, semantic search, and AI engineering.

Skills

- **Programming:** Python, JavaScript, SQL
- **Machine Learning:** Machine Learning, Deep Learning, NLP, Scikit-learn, Pandas, NumPy, Model Evaluation
- **Generative AI:** LLMs, Prompt Engineering, Embeddings, RAG, Fine-tuning, Semantic Search
- **Web Development:** HTML, CSS, React.js, Tailwind CSS, API, GSAP
- **Agentic AI & AI Engineering:** LangChain, LangGraph, Agentic AI / AI Agents, FastAPI, Qdrant, Pydantic, Vector Search
- **Databases:** Vector Databases (Qdrant and others), SQL-based Databases
- **Tools & Cloud:** Git, GitHub, Jupyter Notebook, VS Code, Docker, AWS/Cloud
- **Soft Skills:** Communication, Leadership, Team Collaboration, Problem-Solving

Experience

Research Intern

Jan 2025 – Feb 2025

Modern Education Society's Wadia College of Engineering, Pune

- Worked on HEALTHAI, a machine learning project for Alzheimer's Disease detection using MRI imaging and structured clinical data.
- Performed data cleaning and feature engineering using Python, Pandas, and NumPy.
- Built and evaluated Scikit-learn classification models for CN, MCI, and AD categories, achieving 96%+ accuracy.
- Applied Grad-CAM explainability for model interpretation and feature importance analysis.
- **Technologies:** Python, Pandas, NumPy, Scikit-learn, Machine Learning.

Projects

AI-Powered DSA YouTube Transcript RAG Search & Evaluation System

Python, RAG, Qdrant, SentenceTransformers, Groq, FastAPI

- Built a RAG-based semantic search system over 70+ DSA YouTube video transcripts.
- Implemented transcript chunking, 384-dimensional embeddings, and timestamp-aware retrieval using SentenceTransformers and Qdrant.
- Integrated metadata filtering, payload indexing, and semantic search for efficient context retrieval.
- Integrated Groq LLM and RAG evaluation techniques to improve retrieval relevance and grounded response generation.

AI-Powered Customer Churn Prediction & Retention Assistant

Python, Machine Learning, LLMs, RAG, Qdrant, FastAPI

- Built a machine learning-based customer churn prediction system with LLM-powered explanations.
- Implemented feature engineering, classification, model evaluation, and churn-risk analysis.
- Integrated RAG and agentic workflows for evidence-based customer retention recommendations.
- Used Qdrant and FastAPI for knowledge retrieval and AI application development.

AI-Powered Job Application & Resume Intelligence Platform

Python, LLMs, RAG, Qdrant, FastAPI, Agentic AI

- Developed an AI-powered resume and job-description analysis platform using LLMs and semantic embeddings.
- Implemented resume/JD parsing, semantic skill matching, and skill-gap detection.
- Built Qdrant-based vector search and FastAPI backend services.
- Designed agentic workflows for job-specific interview questions and preparation recommendations.

Education

B.E. Computer Engineering

Modern Education Society's Wadia College of Engineering, Pune

2022 – 2026

SGPA: 7.84/10

Certifications & Activities

1. Sheryians AI Powered Cohort of Web Development
2. Generative AI & Agentic AI by ExcelR
3. Prompt Engineering for Developers